M-finescan: vanilla vs orthogonal AL-RNN

```
Sweep:  M in {10, 11, 14, 15}  x  P in {2, 3, 4, 5, 6}
Epochs: 1000   alpha: 1.0   shared frozen B per M   seed: 42

Metrics:
  rmse_train_final - sqrt(final training MSE), one-step prediction
  rmse_h100        - free-run RMSE on the first 100 steps after t=0
  Dstsp            - KL(p_true || p_gen) on a 30^3 binned 3-D density
  PSE              - mean Hellinger distance between smoothed power spectra
                     (the AL-RNN paper's attractor-quality metrics)
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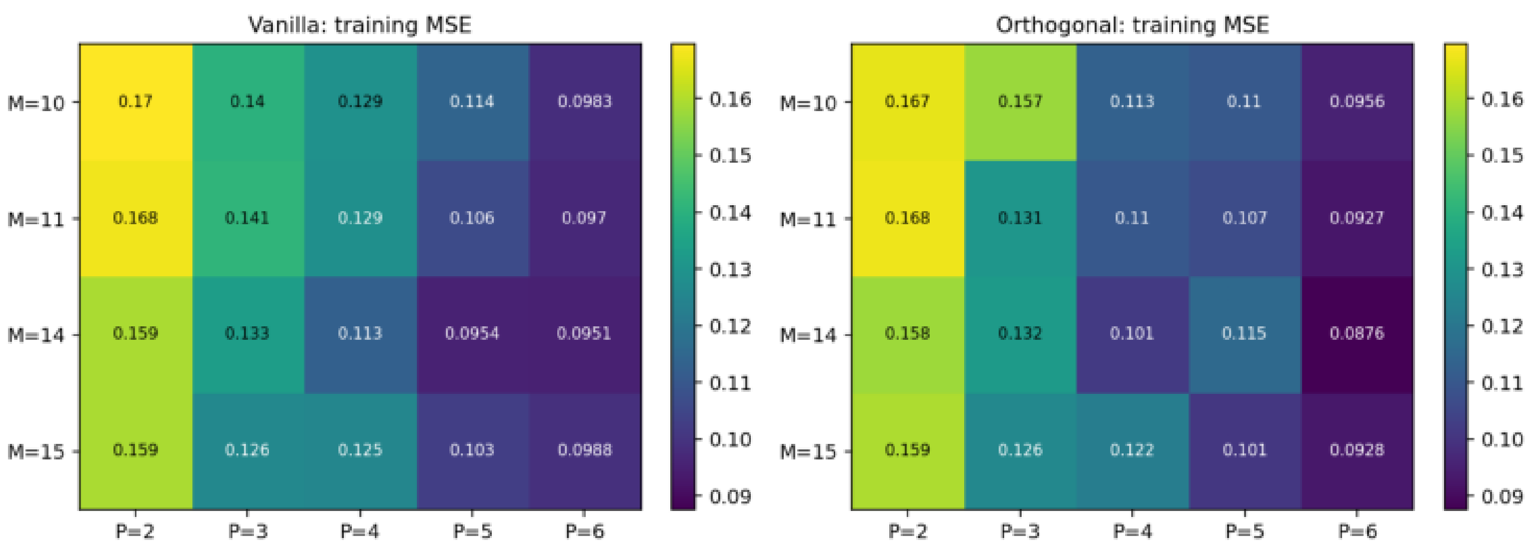
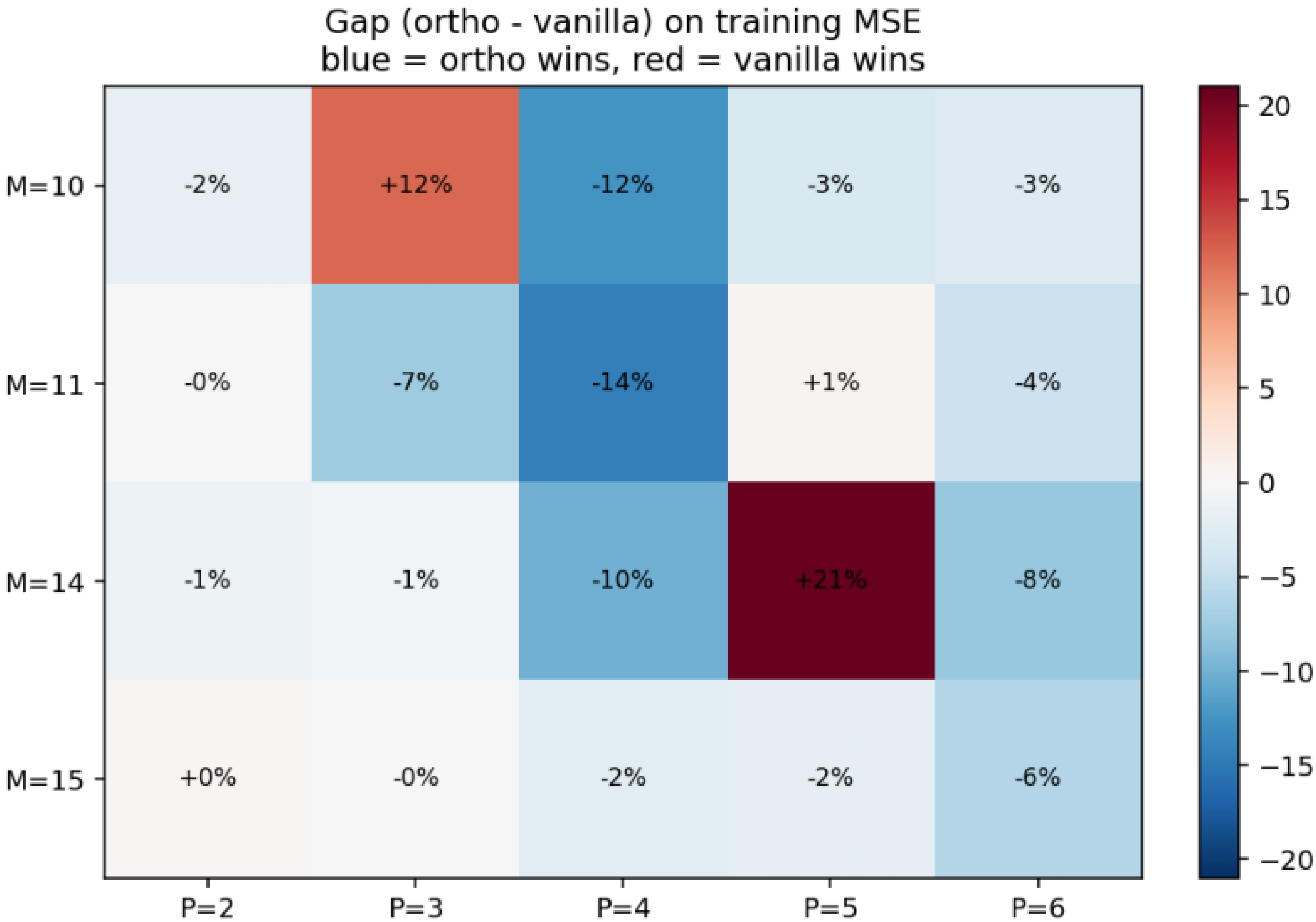
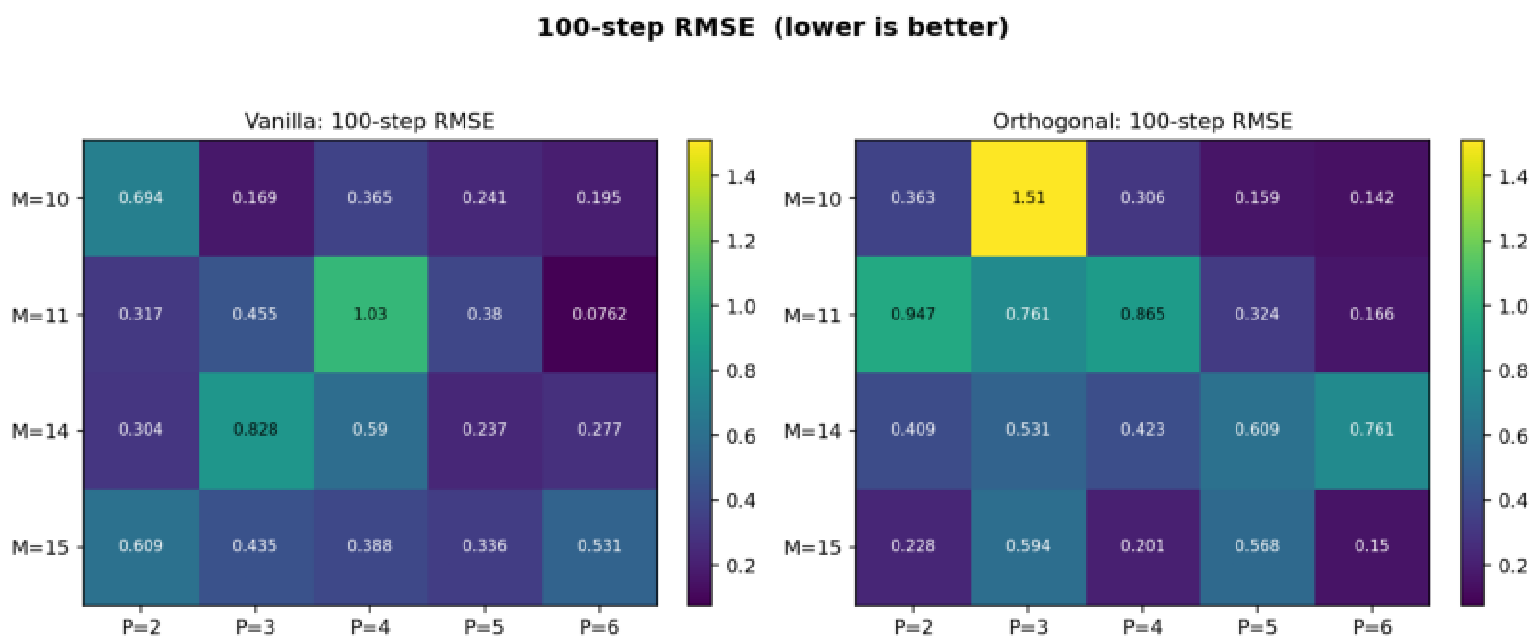
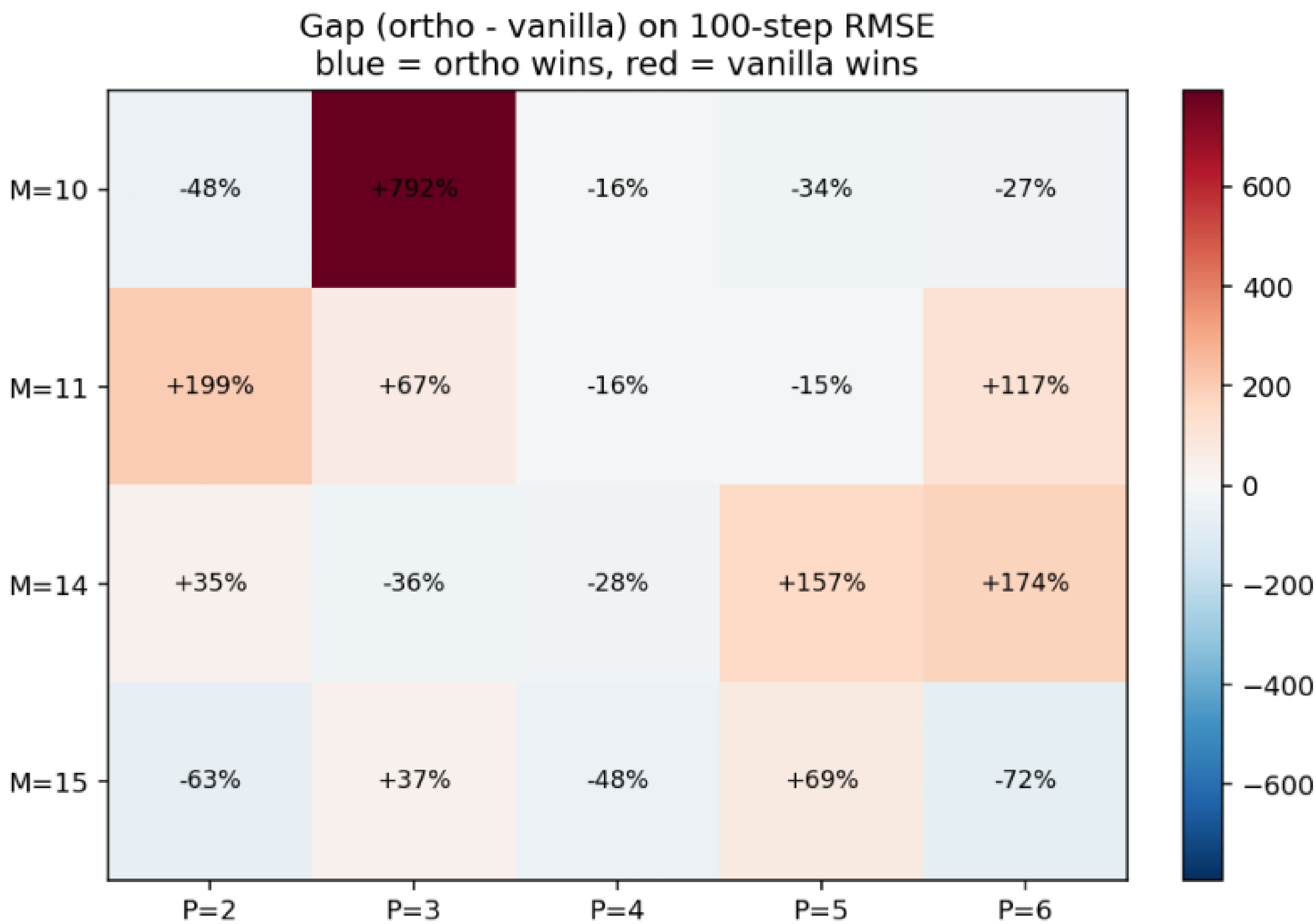
training MSE (lower is better)



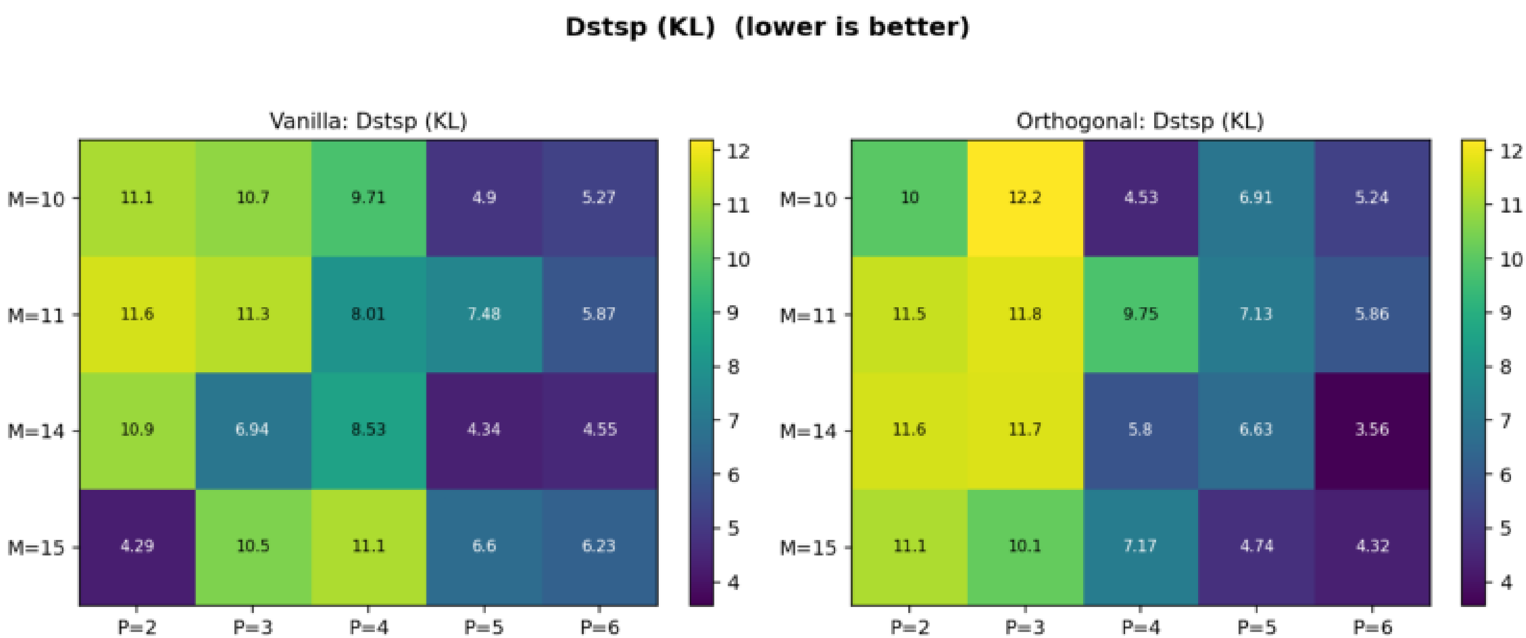


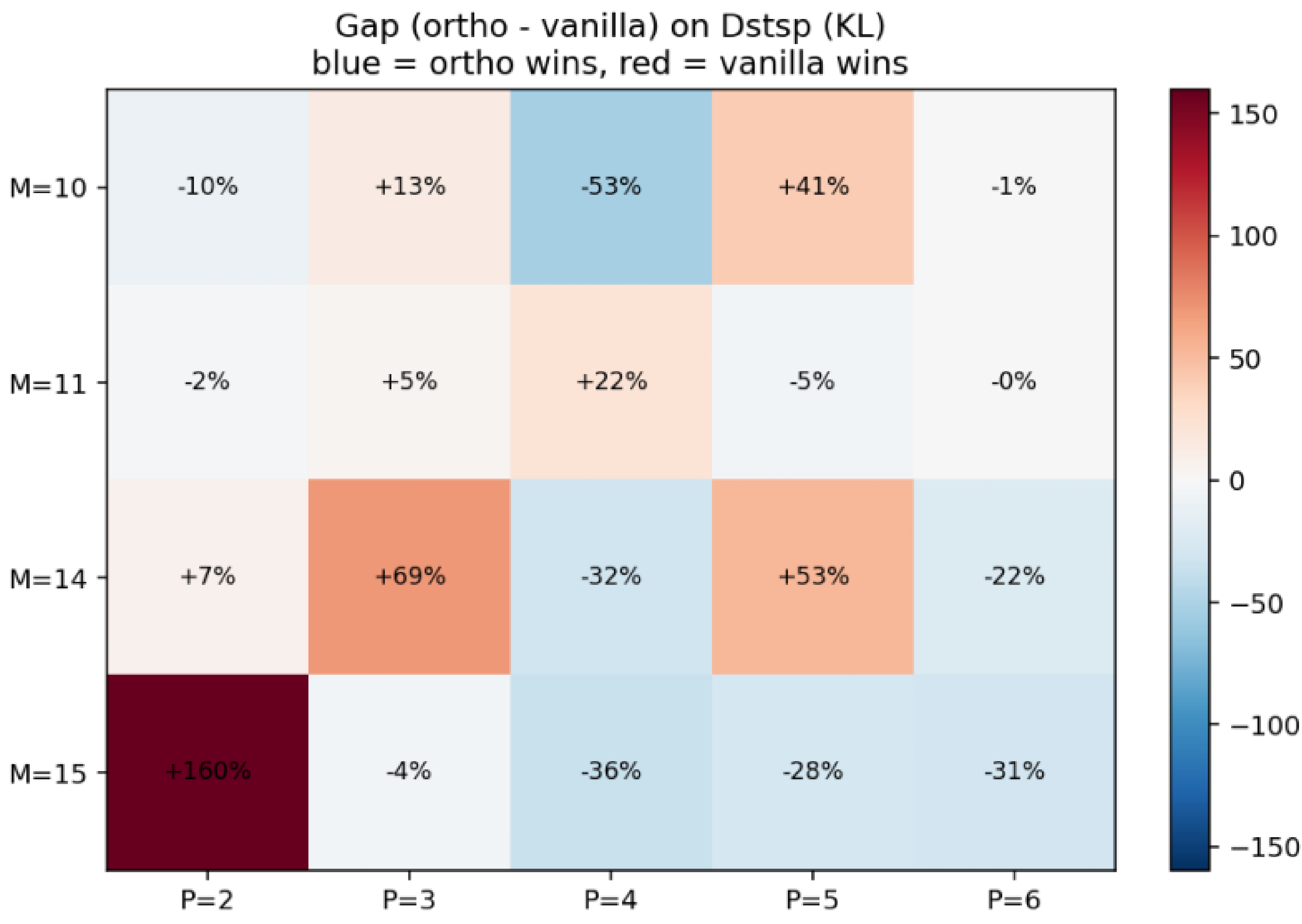
Gap (ortho - vanilla) on training MSE
blue = ortho wins, red = vanilla wins
M=10
M=11
M=14
M=15
P=2
P=3
P=4
P=5
P=6
-2%
+12%
-12%
-3%
-3%
-0%
-7%
-14%
+1%
-4%
-1%
-1%
-10%
+21%
-8%
+0%
-0%
-2%
-2%
-6%
20
15
10
5
0
−5
−10
−15
−20

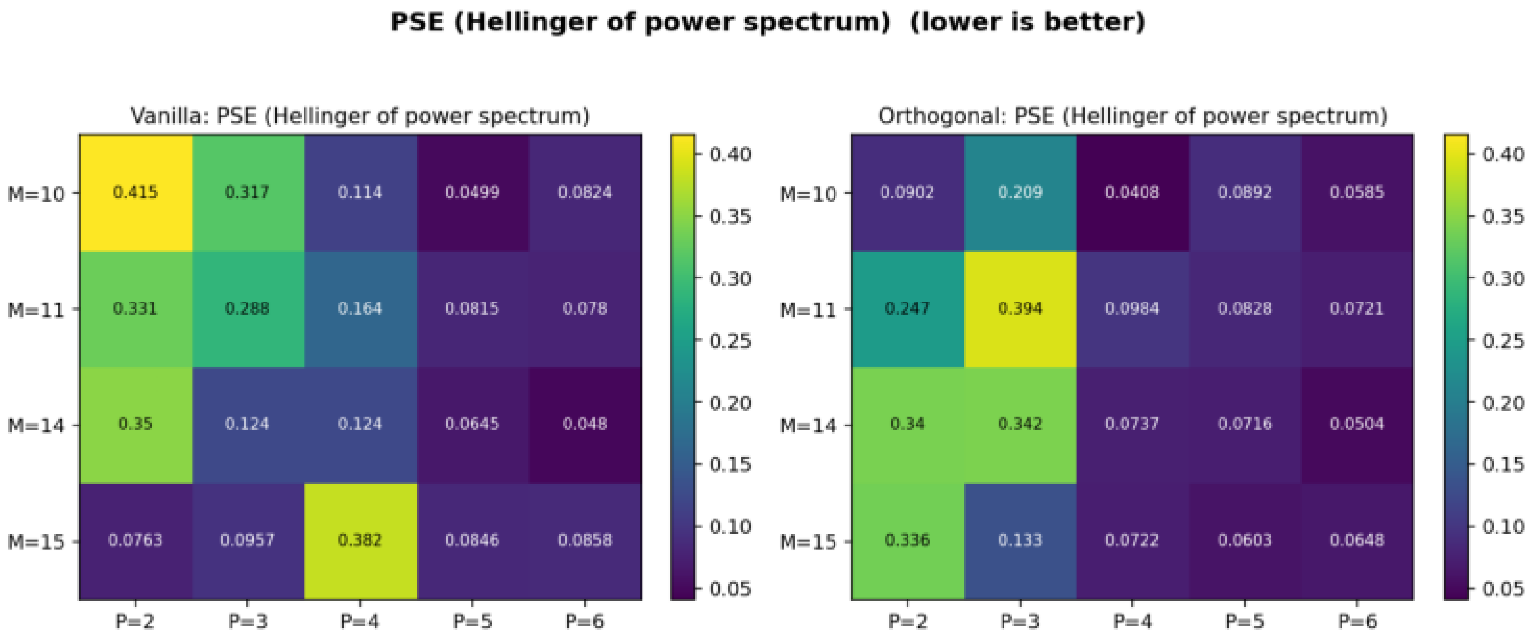

100-step RMSE (lower is better)
Vanilla: 100-step RMSE
M=10
M=11
M=14
M=15
P=2
P=3
P=4
P=5
P=6
0.694 0.169 0.365 0.241 0.195
0.317 0.455 1.03 0.38 0.0762
0.304 0.828 0.59 0.237 0.277
0.609 0.435 0.388 0.336 0.531
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Orthogonal: 100-step RMSE
M=10
M=11
M=14
M=15
P=2
P=3
P=4
P=5
P=6
0.363 1.51 0.306 0.159 0.142
0.947 0.761 0.865 0.324 0.166
0.409 0.531 0.423 0.609 0.761
0.228 0.594 0.201 0.568 0.15
0.2
0.4
0.6
0.8
1.0
1.2
1.4

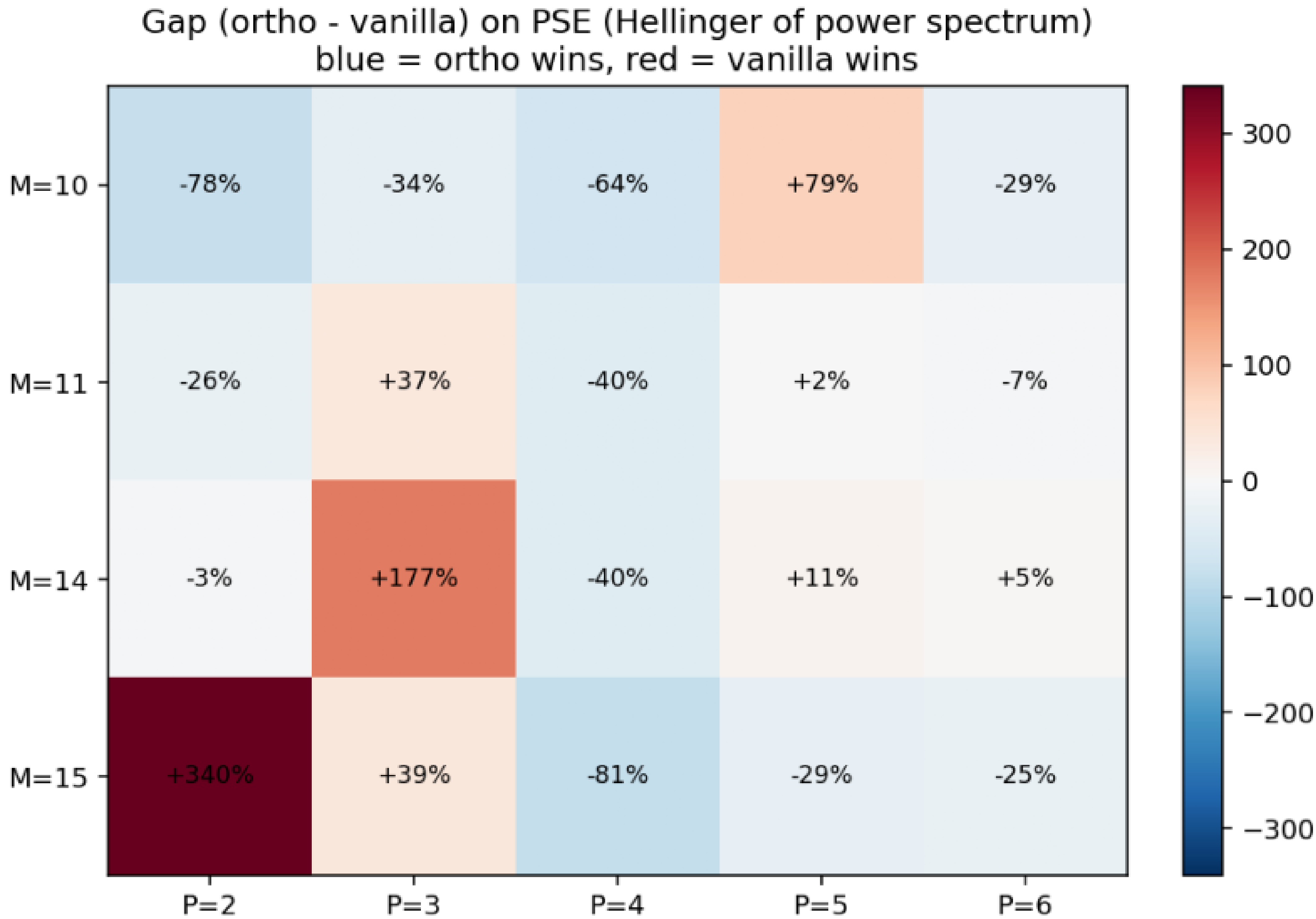


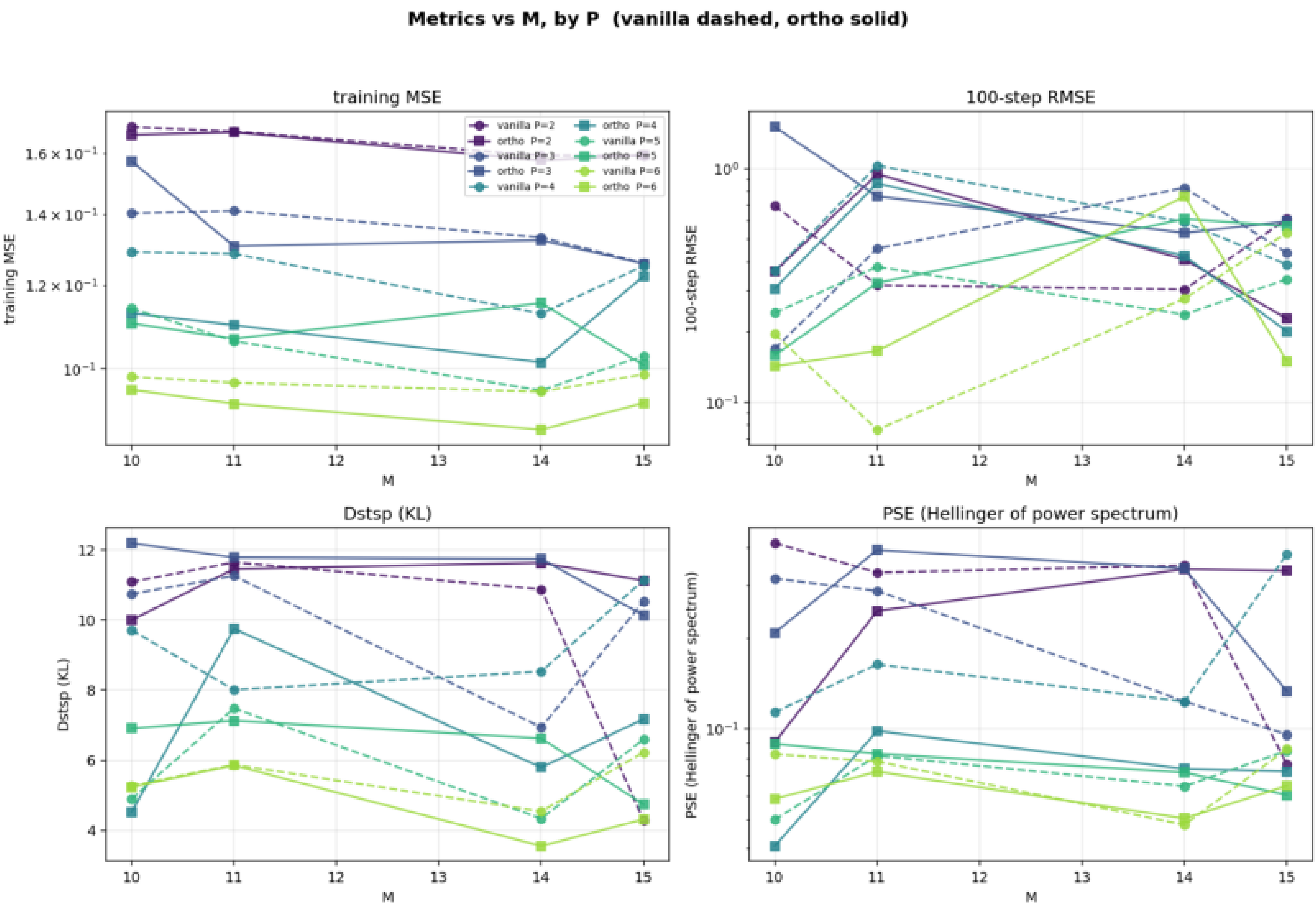
Gap (ortho - vanilla) on 100-step RMSE
blue = ortho wins, red = vanilla wins
M=10
M=11
M=14
M=15
-48%
+792%
-16%
-34%
-27%
+199%
+67%
-16%
-15%
+117%
+35%
-36%
-28%
+157%
+174%
-63%
+37%
-48%
+69%
-72%
P=2
P=3
P=4
P=5
P=6
600
400
200
0
−200
−400
−600


Dstsp (KL) (lower is better)
Vanilla: Dstsp (KL)
M=10
M=11
M=14
M=15
P=2
P=3
P=4
P=5
P=6
11.1 10.7 9.71 4.9 5.27
11.6 11.3 8.01 7.48 5.87
10.9 6.94 8.53 4.34 4.55
4.29 10.5 11.1 6.6 6.23
Orthogonal: Dstsp (KL)
10 12.2 4.53 6.91 5.24
11.5 11.8 9.75 7.13 5.86
11.6 11.7 5.8 6.63 3.56
11.1 10.1 7.17 4.74 4.32
4
5
6
7
8
9
10
11
12


Gap (ortho - vanilla) on Dstsp (KL)
blue = ortho wins, red = vanilla wins
M=10
M=11
M=14
M=15
P=2
P=3
P=4
P=5
P=6
-10%
+13%
-53%
+41%
-1%
-2%
+5%
+22%
-5%
-0%
+7%
+69%
-32%
+53%
-22%
+160%
-4%
-36%
-28%
-31%
150
100
50
0
−50
−100
−150


PSE (Hellinger of power spectrum) (lower is better)
Vanilla: PSE (Hellinger of power spectrum)
M=10
0.415
0.317
0.114
0.0499
0.0824
M=11
0.331
0.288
0.164
0.0815
0.078
M=14
0.35
0.124
0.124
0.0645
0.048
M=15
0.0763
0.0957
0.382
0.0846
0.0858
P=2
P=3
P=4
P=5
P=6
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
Orthogonal: PSE (Hellinger of power spectrum)
M=10
0.0902
0.209
0.0408
0.0892
0.0585
M=11
0.247
0.394
0.0984
0.0828
0.0721
M=14
0.34
0.342
0.0737
0.0716
0.0504
M=15
0.336
0.133
0.0722
0.0603
0.0648
P=2
P=3
P=4
P=5
P=6
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05


Gap (ortho - vanilla) on PSE (Hellinger of power spectrum)
blue = ortho wins, red = vanilla wins
M=10
M=11
M=14
M=15
P=2
P=3
P=4
P=5
P=6
-78%
-34%
-64%
+79%
-29%
-26%
+37%
-40%
+2%
-7%
-3%
+177%
-40%
+11%
+5%
+340%
+39%
-81%
-29%
-25%
300
200
100
0
−100
−200
−300



Metrics vs M, by P (vanilla dashed, ortho solid)
training MSE
100-step RMSE
Dstsp (KL)
PSE (Hellinger of power spectrum)
vanilla P=2
ortho P=2
vanilla P=3
ortho P=3
vanilla P=4
ortho P=4
vanilla P=5
ortho P=5
vanilla P=6
ortho P=6
M